Exchange Rates and the Foreign Exchange Market

Chien-Ping Chung

Preview

- Definitions of Exchange Rates
- Depreciation and Appreciation
- The Foreign Exchange Markets
- Determination of Exchange Rates

Definitions of Exchange Rates

- Exchange rates are quoted as domestic currency per unit of foreign currency or foreign currency per unit of domestic currency.
  - How much can be exchanged for one US dollar? NT$32/US$1
  - How much can be exchanged for one NT dollar? US$0.03125/NT$1
- Exchange rates allow us to denominate the cost or price of a good or service in a common currency.
  - How much does a Ford cost? US$30,000
  - Or US$30,000 x NT$32/US$1 = NT$960,000
  - How much does a Luxgen cost? NT$640,000
  - Or NT$640,000 x US$0.03125/NT$1 = US$20,000

Definitions of Exchange Rates (cont.)

- "Indirect (Quantity) quotation"(or European terms)

  $\Rightarrow$ How many Euros equal 1 Dollar?

  $E_{€/\$} = 0.83215$ $\Rightarrow$ \$ 1 buys 0.83215 €

- "Direct (Price) quotation" (or American terms)

  $\Rightarrow$ How many Dollars equal 1 Euro?

  $E_{\$/€} = 1.2017$ $\Rightarrow$ 1 € buys \$1.2017

Depreciation and Appreciation

- **Depreciation** is a decrease in the value of a currency relative to another currency.
  - A depreciated currency is *less valuable* (less expensive) and therefore can be exchanged for (can buy) a smaller amount of foreign currency.
  - $1/€1 → $1.20/€1 means that the dollar has depreciated relative to the euro. It now takes $1.20 to buy one euro, so that the dollar is less valuable.
  - The euro has appreciated relative to the dollar: it is now more valuable.

Depreciation and Appreciation (cont.)

- **Appreciation** is an increase in the value of a currency relative to another currency.
  - An appreciated currency is *more valuable* (more expensive) and therefore can be exchanged for (can buy) a larger amount of foreign currency.
  - $1/€1 → $0.90/€1 means that the dollar has appreciated relative to the euro. It now takes only $0.90 to buy one euro, so that the dollar is more valuable.
  - The euro has depreciated relative to the dollar: it is now less valuable.

Depreciation and Appreciation (cont.)

- A depreciated currency is less valuable, and therefore it can buy fewer foreign produced goods that are denominated in foreign currency.
  - How much does a Ford cost? US$30,000
  - US$30,000 x NT$32/US$1 = NT$960,000
  - US$30,000 x NT$33/US$1 = NT$990,000
  - How much does a Luxgen cost? NT$640,000
  - NT$640,000 ÷ NT$32/US$1 = US$20,000
  - NT$640,000 ÷ NT$33/US$1 = US$19,394
- A depreciated currency means that imports are more expensive and domestically produced goods and exports are less expensive. Thus, it's favorable to exports, but it's unfavorable to imports.
- A depreciated currency lowers the price of exports relative to the price of imports.

Depreciation and Appreciation (cont.)

- An appreciated currency is more valuable, and therefore it can buy more foreign produced goods that are denominated in foreign currency.
  - How much does a Ford cost? US$30,000
  - US$30,000 x NT$32/US$1 = NT$960,000
  - US$30,000 x NT$30/US$1 = NT$900,000
  - How much does a Luxgen cost? NT$640,000
  - NT$640,000 ÷ NT$32/US$1 = US$20,000
  - NT$640,000 ÷ NT$30/US$1 = US$21,333
- An appreciated currency means that imports are less expensive and domestically produced goods and exports are more expensive. Thus, it's favorable to imports, but it's unfavorable to exports.
- An appreciated currency raises the price of exports relative to the price of imports.

Foreign Exchange Markets

- The set of markets where foreign currencies and other assets are exchanged for domestic ones
  - Institutions buy and sell deposits of currencies or other assets for investment purposes.
- The market in which the currency of one country is exchanged for the currency of another.
  - The foreign exchange market that is made up of importers and exporters, banks, and specialist dealers who buy and sell currencies.

Foreign Exchange Markets (cont.)

The participants:

1. **Commercial banks and other depository institutions**: transactions involve buying/selling of deposits in different currencies for investment purposes.
2. **Non-bank financial institutions** (mutual funds, hedge funds, securities firms, insurance companies, pension funds) may buy/sell foreign assets for investment.
3. **Non-financial businesses conduct foreign currency transactions** to buy/sell goods, services and assets.
4. **Central banks**: conduct official international reserves transactions.

Determination of Exchange Rates

- The value of the foreign exchange rate fluctuates.
- Sometimes the domestic currency depreciates and sometimes it appreciates. Why?
- The foreign exchange rate is a price and like all prices, demand and supply in the foreign exchange market determine its value.
- In this section, we assume U.S. dollars is a foreign currency, Japanese yen is a domestic currency.

Determination of Exchange Rates (cont.)

- Demand in the Foreign Exchange Market
  - The quantity of dollars demanded in the foreign exchange market is the amount that traders plan to buy during a given period at a given exchange rate.
  - The demand curve shows how exchange rate affects quantity of dollars demanded, *other things being equal*.
  - The quantity of dollars demanded depends on: the exchange rate

Determination of Exchange Rates (cont.)

- The Law of Demand for Foreign Exchange
  - Other things remaining the same, the higher the exchange rate, the smaller is the quantity of dollars demanded.
  - The exchange rate influences the quantity of dollars demanded for:
    - Imports effect for Japan (Exports effect for U.S.)

Determination of Exchange Rates (cont.)

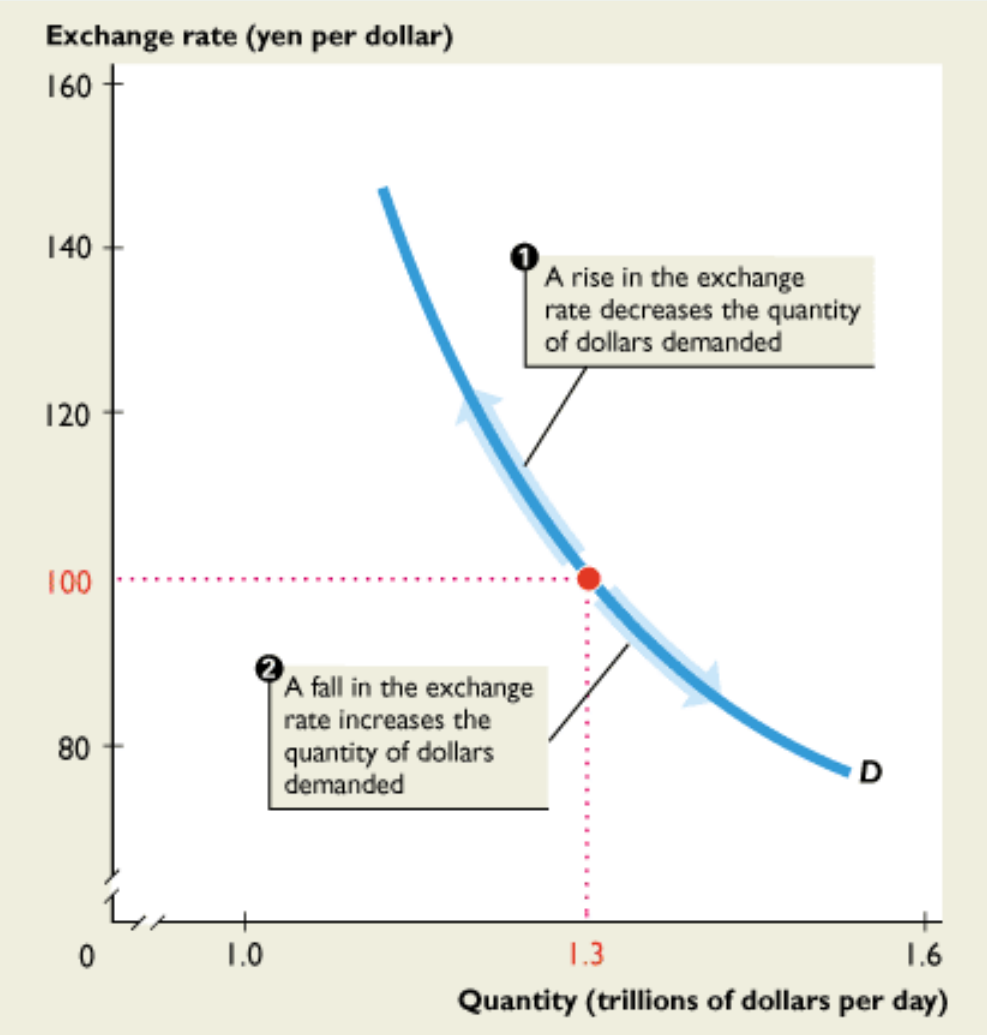
- **Imports Effect for Japan(Exports Effect for U.S.)**
- The larger the value of U.S. exports, the larger is the quantity of dollars demanded on the foreign exchange market in Japan.
- The lower the exchange rate, the cheaper are U.S.-made goods and services to people in Japan, the more the United States exports, and the greater is the quantity of U.S. dollars demanded to pay for them.

Determination of Exchange Rates (cont.)

Figure 1 shows the demand for dollars.

1. If the exchange rate rises, the quantity of dollars demanded decreases along the demand curve for dollars.
2. If the exchange rate falls, the quantity of dollars demanded increases along the demand curve for dollars.

Figure 1



Determination of Exchange Rates (cont.)

- Changes in the Demand for Dollars
  - ♦ Any influence that changes the quantity of U.S. dollars that people plan to buy in the foreign exchange market other than the exchange rate changes the demand for U.S. dollars and shifts the demand curve for dollars.
  - ♦ These influences are:
    - Interest rates in the United States and other countries
    - Expected future exchange rate

Determination of Exchange Rates (cont.)

- **Interest Rates in the United States and Japan**
- **U.S. interest rate differential**
- The U.S. interest rate minus the Japan interest rate.
- Other things remaining the same, the larger the U.S. interest rate differential, the greater is the demand for U.S. assets and the greater is the demand for dollars on the foreign exchange market.

Determination of Exchange Rates (cont.)

- **The Expected Future Exchange Rate**
- Other things remaining the same, the higher the expected future exchange rate, the greater is the demand for dollars.
- The higher the expected future exchange rate, the larger is the expected profit from holding dollars, so the larger is the quantity of dollars that people plan to buy on the foreign exchange market.

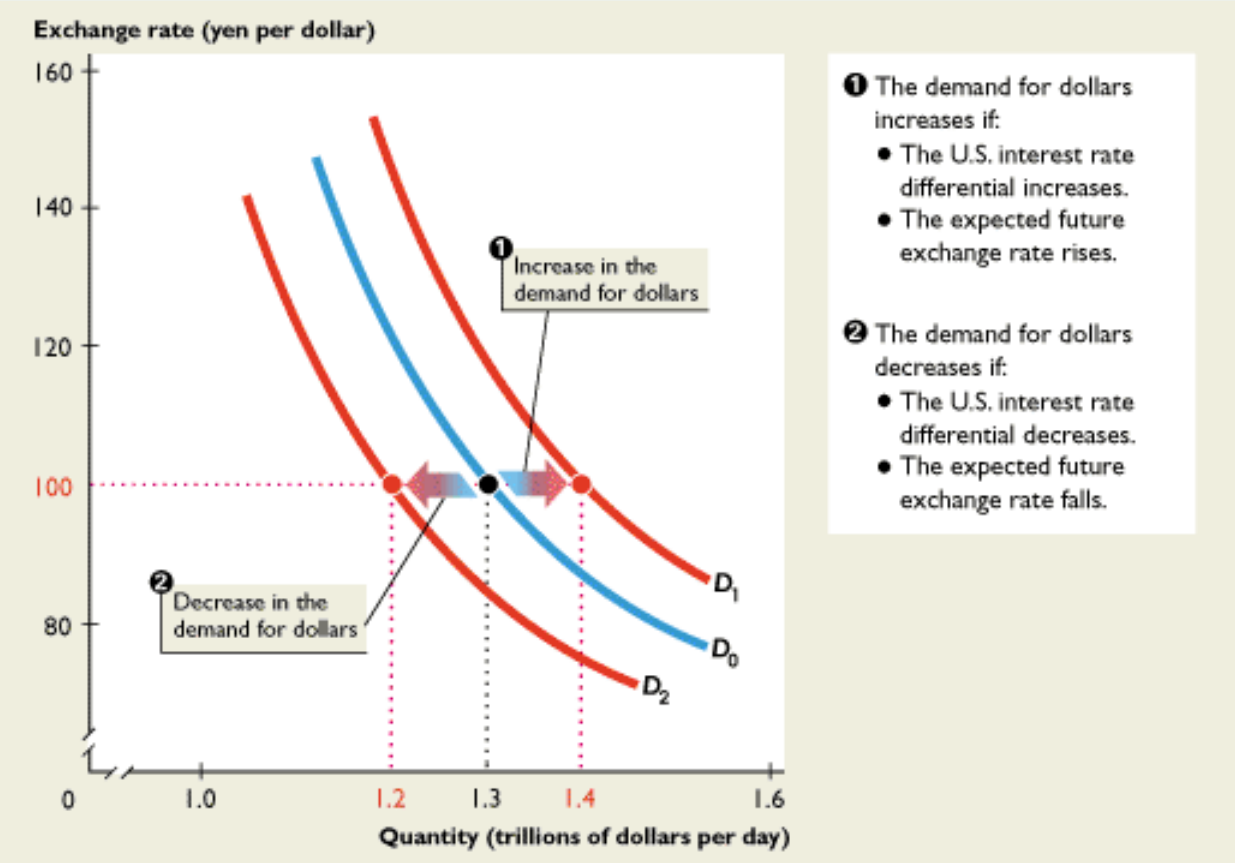
Determination of Exchange Rates (cont.)

Figure 2 shows changes in the demand for dollars.

1. An increase in the demand for dollars

2. A decrease in the demand for dollars.

Figure 2



Determination of Exchange Rates (cont.)

- Supply in the Foreign Exchange Market
  - The quantity of U.S dollars supplied in the foreign exchange market is the amount that traders plan to sell during a given time period at a given exchange rate.
  - The supply curve shows how exchange rate affects quantity of U.S dollars supplied, *other things being equal*.
  - The quantity of U.S. dollars supplied depends on: the exchange rate

Determination of Exchange Rates (cont.)

- The Law of Supply of Foreign Exchange
  - Traders supply U.S. dollars in the foreign exchange market when they buy other currencies.
  - Other things remaining the same, the higher the exchange rate, the greater is the quantity of U.S. dollars supplied in the foreign exchange market.
  - The exchange rate influences the quantity of dollars supplied for:
    - Exports effect for Japan (Imports effect for U.S.)

Determination of Exchange Rates (cont.)

- **Exports Effect for Japan(Imports Effect for U.S.)**
- The larger the value of U.S. imports, the larger is the quantity of foreign currency demanded to pay for these imports.
- When people buy foreign currency, they supply dollars.
- Other things remaining the same, the higher the exchange rate, the cheaper are foreign-made goods and services to Americans. So the more the United States imports, the greater is the quantity of U.S. dollars supplied on the foreign exchange market.

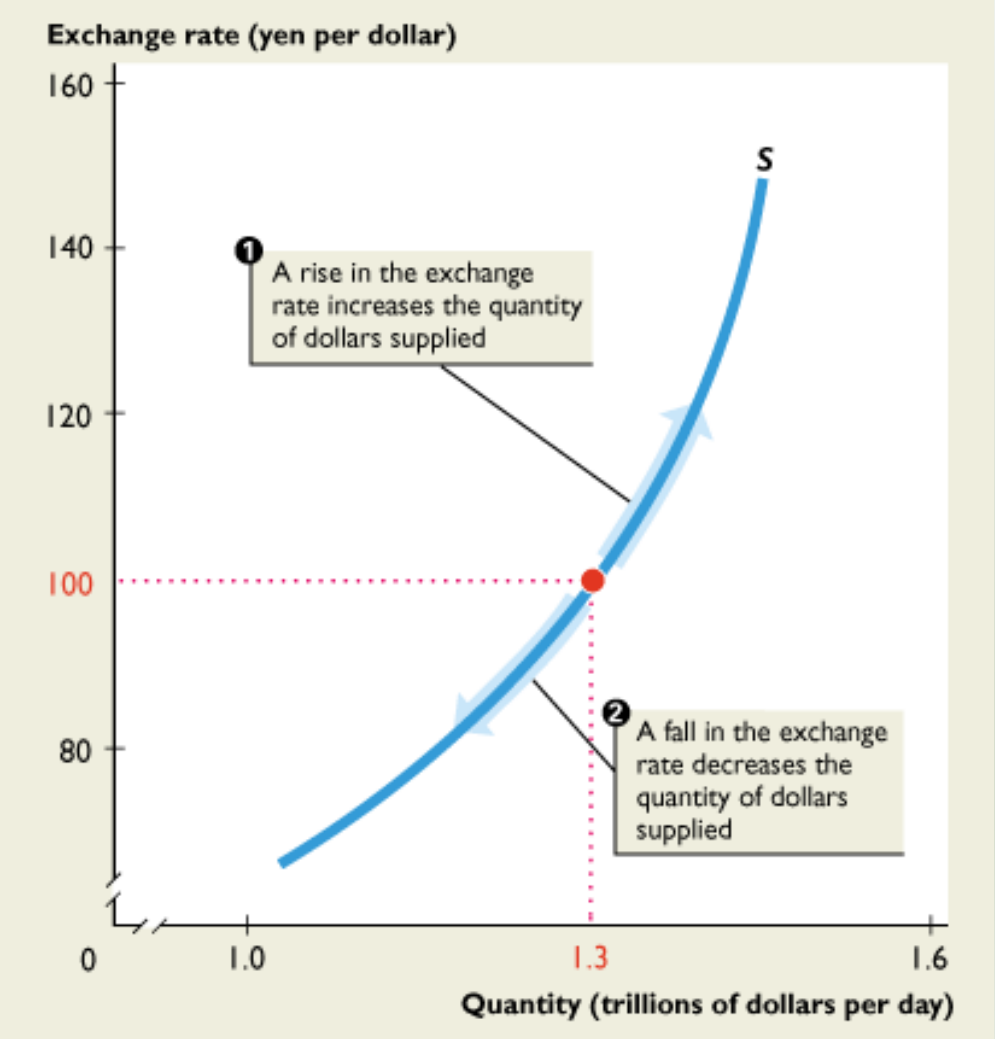
Determination of Exchange Rates (cont.)

Figure 3 shows the supply of dollars.

1. If the exchange rate rises, the quantity of dollars supplied increases along the supply curve for dollars.

2. If the exchange rate falls, the quantity of dollars supplied decreases along the supply curve for dollars.

Figure 3



Determination of Exchange Rates (cont.)

- Changes in the Supply of Dollars
  - ♦ Any influence (other than the current exchange rate) that changes the quantity of U.S. dollars that people plan to sell in the foreign exchange market changes the supply of U.S. dollars and shifts the supply curve for dollars.
  - ♦ These influences are:
    - Interest rates in the United States and Japan
    - Expected future exchange rate

Determination of Exchange Rates (cont.)

- **Interest Rates in the United States and Other Countries**
- The larger the U.S. interest rate differential, the smaller is the demand for foreign assets and so the smaller is the supply of U.S. dollars on the foreign exchange market.

Determination of Exchange Rates (cont.)

- **The Expected Future Exchange Rate**
- Other things remaining the same, the higher the expected future exchange rate, the smaller is the expected profit from selling U.S. dollars today, so the smaller is the supply of dollars today.

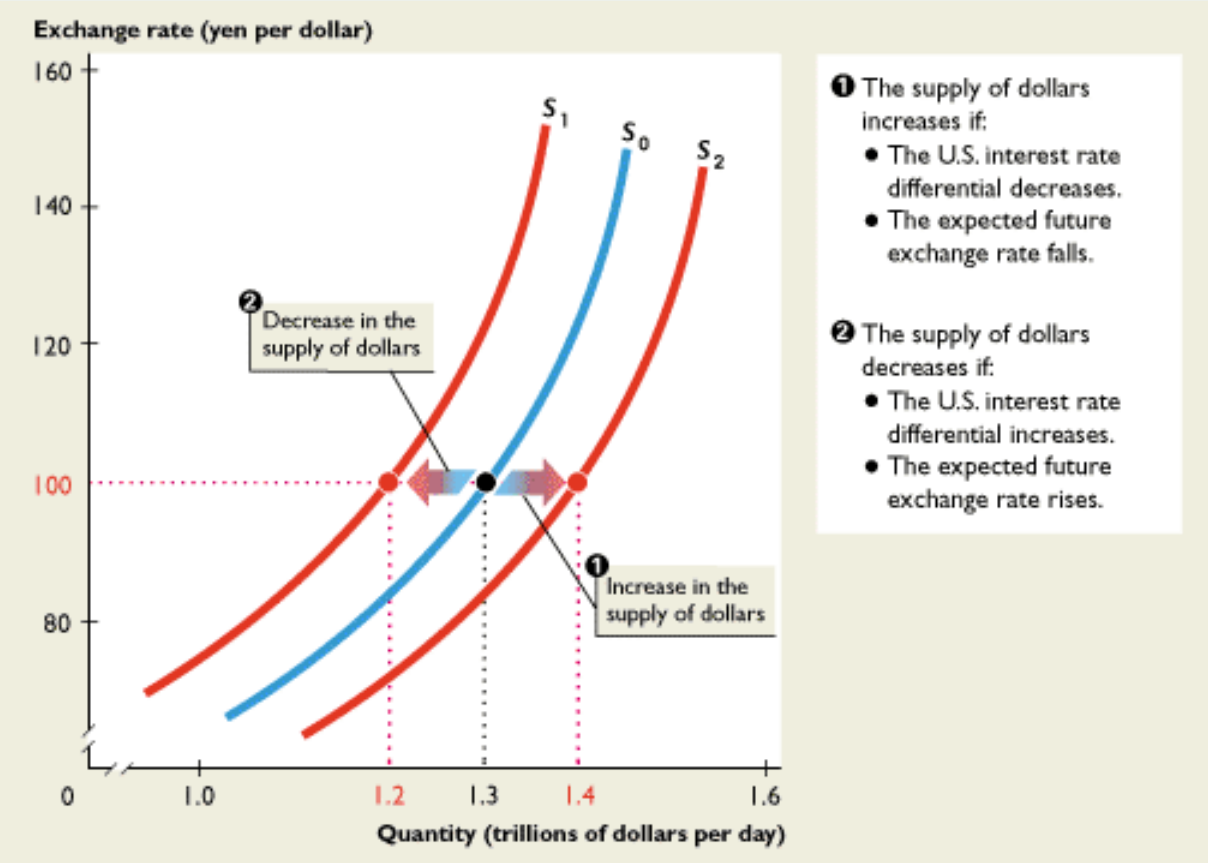
Determination of Exchange Rates (cont.)

Figure 4 shows changes in the supply of dollars.

1. An increase in the supply of dollars.

2. A decrease in the supply of dollars.

Figure 4



Determination of Exchange Rates (cont.)

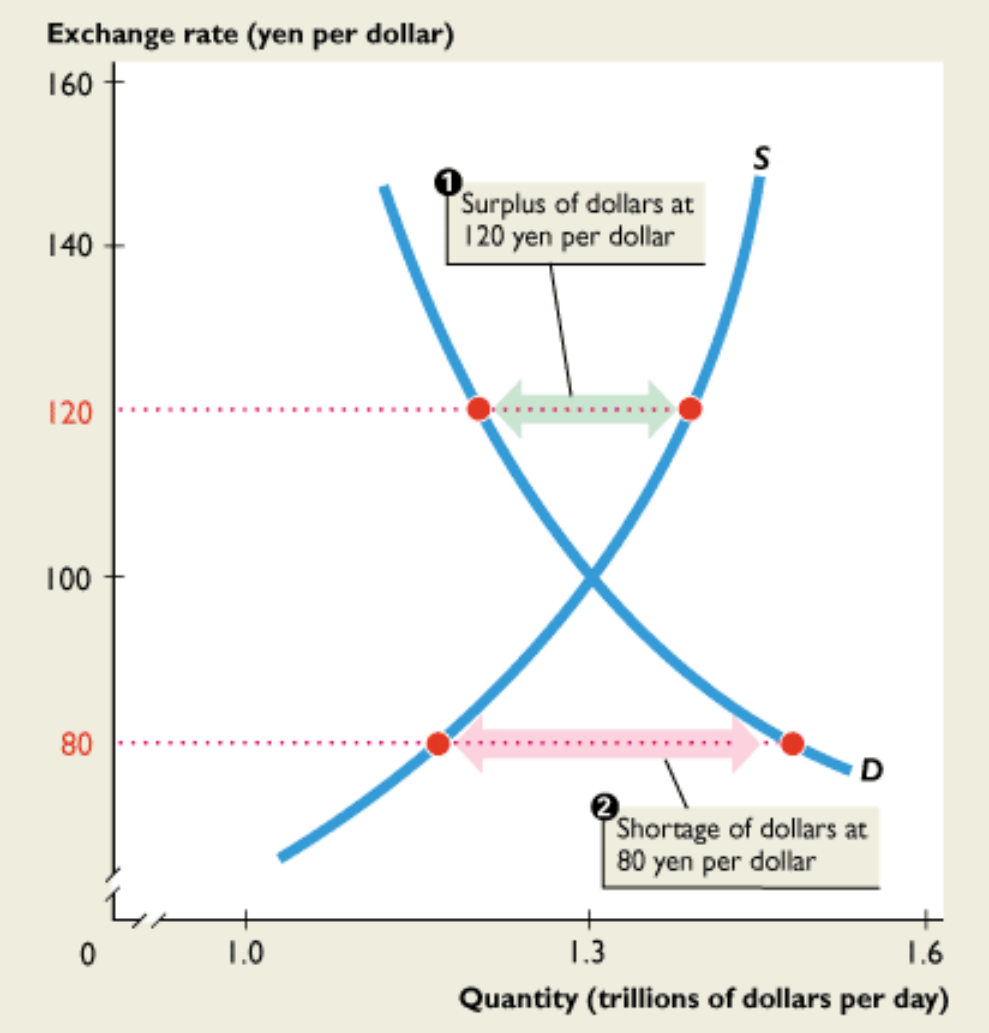
- Market Equilibrium
  - Demand and supply in the foreign exchange market determines the exchange rate.
  - If the exchange rate is too low, there is a shortage of dollars.
  - If the exchange rate is too high, there is a surplus of dollars.
  - At the equilibrium exchange rate, there is neither a shortage nor a surplus.

Determination of Exchange Rates (cont.)

Figure 5

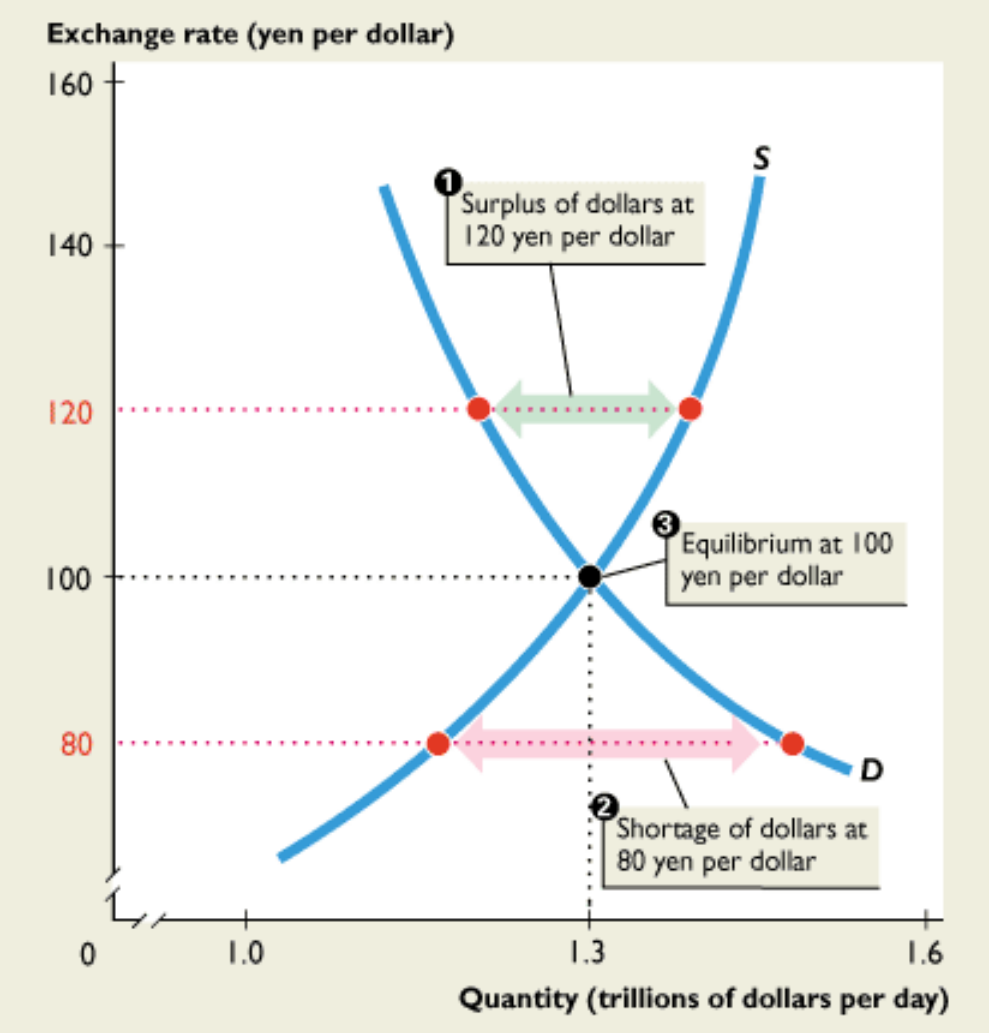
Figure 5 shows the equilibrium exchange rate.

1. If the exchange rate is 120 yen per dollar, there is a surplus of dollars and the exchange rate falls.
2. If the exchange rate is 80 yen per dollar, there is a shortage of dollars and the exchange rate rises.



Determination of Exchange Rates (cont.)

3. If the exchange rate is 100 yen per dollar, there is neither a shortage nor a surplus of dollars and the exchange rate remains constant.The market is in equilibrium.



Determination of Exchange Rates (cont.)

- Changes in the Exchange Rate
  - The predictions about the effects of changes in the demand for and supply of dollars are exactly the same as for any other market.
  - An increase in the demand for dollars with no change in supply raises the exchange rate.
  - A increase in the supply of dollars with no change in demand lowers the exchange rate.

Determination of Exchange Rates (cont.)

- Three Steps to Analyzing Changes in Eq'm ***Exchange Rate:***

To determine the effects of any event,

1. Decide whether event shifts ***S*** curve, ***D*** curve, or both.
2. Decide in which direction curve shifts.
3. Use supply-demand diagram in the foreign exchange market to see how the shift changes eq'm ***exchange rate***.

ACTIVE LEARNING
Shifts in supply and demand curve in the foreign exchange market

- Please answer following questions (Using demand and supply curve in the foreign exchange market determine its value)
- (1) A rise in the U.S. dollar interest rate how to affect NT$/US$ exchange rate?
- (2) A fall in the expected future exchange rate how to affect NT$/US$ exchange rate?